ADEMP protocol: paper 02 (carryover-sensitivity)

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Manuscript. `analysis/report/02-carryover-sensitivity/report.Rmd`

Protocol history. Original grid plan dated 2026-04-15; reformatted into ADEMP structure 2026-05-08.

This document is the working protocol for the exploratory simulation study described in the carryover-sensitivity manuscript (paper 02), written following the ADEMP reporting framework of @morris2019simulation. The study is exploratory, and its findings should be confirmed in an independent study. The protocol restates the parameter grid, performance measures, and replicate budget specified in `simulation-grid-plan.md` (the original design document) in the formal Aims / Data-generating-mechanism / Estimands / Methods / Performance-measures structure SIM editors expect. Changes to the working protocol are recorded at the foot of this document.

A. Aims

A1 (principal aim). Determine whether the ranking of analysis-model carryover specifications (A1 binary, A2 exposure-weighted, A3 lagged-nuisance) for detecting the biomarker-treatment interaction in aggregated N-of-1 trials depends on (i) the true data-generating carryover decay form, (ii) the data-generating-process architecture (mean moderation versus multivariate-normal differential correlation), and (iii) the carryover half-life, across the trial-design and sample-size settings used in precision-medicine N-of-1 work.

A2 (sensitivity aims). Evaluate the robustness of the A1 ranking to (S1) within-factor autocorrelation strength, (S2) analyst-versus-truth half-life mismatch, (S3) informative and non-informative dropout, (S4) the full effect-size curve in the biomarker moderation parameter, and (S5) interactions between autocorrelation strength and carryover half-life.

A3 (secondary aim). Quantify the absolute power penalty of analysis-model carryover mis-specification under each DGP architecture. The A3 outputs feed sample-size guidance in the Discussion.

D. Data-generating mechanisms

The simulation generates aggregated N-of-1 trial data using the pmsimstats-ng `tidyverse` simulation collection (`implementations/tidyverse/`). The data-generating model has four conceptual components:

- **Trial design.** One of `CO` (traditional crossover), `Hybrid` (Hendrickson hybrid), and `OL+BDC` (open-label with blinded discontinuation), built by `buildtrialdesign()`.
- **Response trajectory.** Three modGompertz components (BR pharmacological, TV time-variant, PB placebo-belief), parameterised at the prazosin/PTSD reference cell.
- **Biomarker moderation.** Either Architecture A (direct mean-moderation: post-draw shift on on-drug BR scaled by the centred biomarker) or Architecture B (MVN differential correlation: $\mathrm{Cor}(BM, BR)$ on-drug equals c_{bm}; off-drug decays as $c_{bm} \exp(-\lambda t_{sd})$).
- **Carryover decay.** One of linear, exponential, or Weibull with shape parameter $k \in \{0.7, 1.0, 1.5\}$.

Tier 1 principal factorial (540 cells)

Every combination of the seven axes below is evaluated:

Axis	Values	Levels
DGP carryover form	linear, exponential, Weibull($k \in \{0.7, 1.0, 1.5\}$)	5
Analysis specification	A1 binary, A2 exposure-weighted, A3 lagged	3
DGP architecture	mean moderation, MVN differential correlation	2
Carryover half-life $t_{1/2}$ (weeks)	0, 0.5, 1.0	3
Trial design	CO, Hybrid, OL+BDC	3
Sample size N	35, 70	2
Biomarker moderation c_{bm}	0, 0.30, 0.45	3

Cell count: $5 \times 3 \times 2 \times 3 \times 3 \times 2 \times 3 = 540$.

Tier 2 sensitivity blocks (5 blocks, 31 cells)

Each block varies one additional axis at the **reference cell** (exponential, $k = 1.0$, MVN, $t_{1/2} = 1.0$, Hybrid, $N = 70$, $c_{bm} = 0.45$) crossed with the three analysis specifications.

- **S1 autocorrelation.** $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. 4 cells.

- **S2 half-life mismatch.** $t_{1/2}^{\text{true}} \in \{0.5, 1.0\} \times t_{1/2}^{\text{analyst}} \in \{0.25, 0.5, 1.0, 2.0\}$. 8 cells.
- **S3 dropout.** Rate $d \in \{0, 0.1, 0.2, 0.3\}$ × mechanism {MCAR, MAR-by-baseline-severity}, with the $(d = 0, \text{MAR})$ cell collapsed to MCAR. 7 cells.
- **S4 effect-size curve.** $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. 6 cells. ($c_{bm} = 0.60$ feasible only under Architecture A; reported with note.)
- **S5 rho × carryover (exploratory).** $\rho \in \{0.5, 0.8\} \times t_{1/2} \in \{0, 0.5, 1.0\}$. 6 cells.

Tier 1 and Tier 2 share an anchor at the reference cell; results are reported separately and not pooled. The reasoning for the two-tier separation is documented in `simulation-grid-plan.md` §"Relationship between Tier 1 and Tier 2".

E. Estimands

The principal target of inference is the biomarker-by-treatment interaction coefficient $\beta_{bm:D_{bc}}$ in the linear-mixed analysis model $Sx \sim bm + t + D_{bc} + bm{:}D_{bc}$, with random intercept per participant and continuous-time AR(1) residual correlation (`corCAR1(form = ~ t | ptID)`).

Three estimands are evaluated for each cell:

- **E1: Power.** The rejection probability for the $bm{:}D_{bc}$ Wald t-test at $\alpha = 0.05$ under the alternative ($c_{bm} > 0$).
- **E2: Type I error.** The rejection probability under the null ($c_{bm} = 0$). Reported on the `c_bm = 0` cells in Tier 1 and Tier 2 blocks where applicable.
- **E3: Estimator bias.** The mean of $\hat{\beta}_{bm:D_{bc}}$ minus the implied true value of the interaction coefficient under each architecture. Architecture A induces a true value of $c_{bm} \cdot \sigma_{BR}/\sigma_{bm}$; Architecture B induces a true value derivable from the Sigma matrix's $\partial \text{Cor}(BM, BR)/\partial D_{bc}$.

Secondary estimands include the empirical standard deviation of $\hat{\beta}_{bm:D_{bc}}$ across replicates, the median absolute residual of the model fit, and the convergence rate of the `nlme::lme` optimiser.

M. Methods

Three estimators are compared at every cell:

- **A1 binary.** Fit the linear-mixed model with `Db` (binary on-drug indicator: 1 on-drug, 0 off-drug). Strict-RM-ANOVA analogue.

- **A2 matched.** Fit the linear-mixed model with `Dbc` (continuous-decay drug indicator: 1 on-drug, $\exp(-\lambda^{\text{analyst}} t_{sd})$ off-drug, where λ^{analyst} matches the DGP except in S2). pmsimstats-ng default.
- **A3 lagged.** Fit the linear-mixed model with two indicators: the binary on-drug `Db` and a lagged-on indicator that equals 1 at the first off-drug timepoint after each on-drug episode and 0 elsewhere. The lagged indicator absorbs short-range carryover without committing to a parametric decay form [@joneskenward2014].

All three fits use `corCAR1(form = ~ t | ptID)` for the residual correlation structure, random intercept per participant, REML estimation, and the `nlme::lme optim` optimiser with default tolerances. Convergence is recorded for each replicate; failed fits are excluded from the performance summaries (with the counts reported).

P. Performance measures

For every cell, the following are reported with their Monte Carlo standard errors (MCSE):

Performance measure	Formula	MCSE formula
Power	$\hat{\pi} = \frac{1}{n}\sum I(p_i < 0.05)$	$\sqrt{\hat{\pi}(1-\hat{\pi})/n}$
Type I error (null cells)	as above with $c_{bm} = 0$	as above
Mean estimate	$\bar{\beta} = \frac{1}{n}\sum \hat{\beta}_i$	$s_\beta/\sqrt{n}$
Empirical SE	$s_\beta = \text{sd}(\hat{\beta}_i)$	$s_\beta/\sqrt{2(n-1)}$
Bias	$\bar{\beta} - \beta_{\text{true}}$	$s_\beta/\sqrt{n}$
Coverage	$\frac{1}{n}\sum I(\beta_{\text{true}} \in \text{CI}_i)$	$\sqrt{\hat{c}(1-\hat{c})/n}$
Convergence rate	$\frac{1}{n}\sum I(\text{conv}_i = 1)$	$\sqrt{\hat{c}(1-\hat{c})/n}$

For paired cross-specification comparisons within a cell (A2 versus A1, A2 versus A3), McNemar χ^2 tests on the per-replicate rejection indicators are reported with their p-values; this leverages the variance reduction from applying all three specifications to the same simulated datasets.

Replication budget and justification

The principal factorial uses **500 replicates per cell**, giving MCSE on a power estimate near 0.5 of approximately 0.022. At the manuscript's headline cells (power approximately 0.86) the MCSE is approximately 0.015, so the A2-vs-A1 separation of approximately 10 percentage points is approximately five MCSE of separation, sufficient for unambiguous discrimination. The paired McNemar comparisons leverage the within-replicate correlation among A1, A2, and A3

fits, reducing the effective SE of a within-cell ranking comparison below the marginal binomial MCSE.

A high-precision rerun at 600-1000 replicates per cell is specified for a 24-cell slice (3 specs $\times$ 2 c_{bm} $\times$ 2 N $\times$ 2 $t_{1/2}^{\text{DGP}}$) to confirm the headline ranking with MCSE near 0.018, reported as a Prototype Monte Carlo confirmation in the manuscript Discussion section.

Convergence and failure reporting

The `nlme::lme` fit at each replicate produces a `conv` flag. Replicates with conv $= 0$ are counted but excluded from the rejection-rate and bias estimators. Per-cell convergence rates are reported alongside the performance measures. Cells with convergence rate below 90% are flagged in the manuscript Results section and discussed in the Discussion.

In the production run as of 2026-05-08, convergence is $\geq 99.5\%$ across the 540 Tier-1 cells.

Software, versions, and seeds

- R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`).
- Package versions captured in `renv.lock`. Principal packages: `nlme`, `data.table`, `furrr`, `purrr`, `parallel`, `rmarkdown`, `bookdown`.
- Master RNG seed: `set.seed(20260415)` at the top of the driver `01-run-factorial.R`. Per-replicate seeds derive from the master seed by `+ rep_idx`. Tier 2 sensitivity blocks use the same convention.

Planned outputs

Output	Path
Tier 1 per-replicate	`analysis/scripts/carryover-sensitivity/output/01-factorial`
Tier 2 per-replicate	`analysis/scripts/carryover-sensitivity/output/04-sensitivi`
Tier 1 cell summary	`analysis/data/02-grid-summary.rds`
Tier 2 cell summary	`analysis/data/02-sensitivity-summary.rds`
High-precision rerun	`analysis/data/quick-sim/02-carryover-replicates.rds`
Prototype summary	`analysis/data/quick-sim/02-carryover-summary.txt`
Tier 1 figures	`analysis/figures/02-power-by-spec.pdf`, `02-heatmap-matched-vs-mismatched.pdf`, `02-type1-boxplots.pdf`
Tier 2 figures	`analysis/figures/02-sens-S{1..5}.pdf`

Scope exclusions

The following axes are deliberately excluded from this protocol and are noted in the Discussion section of the manuscript:

- **Washout period length and timepoint density.** Design-level mitigation is orthogonal to the analysis-specification question evaluated here. A dedicated paper on design-level mitigation is in the research portfolio.
- **Gompertz response trajectory parameters (max, rate, disp).** Out of scope; the prazosin/PTSD reference parameters are held fixed.
- **Random slopes versus intercept-only random effects.** Orthogonal to carryover specification and would double the grid; a brief note in the manuscript Discussion points to the Hendrickson 2020 sensitivity analysis that addressed this.
- **Multiple-testing correction.** Each simulation tests a single fixed biomarker.
- **Alpha level.** Type I error is evaluated at the standard $\alpha = 0.05$; a supplementary check at $\alpha = 0.10$ is available post hoc.

Changes to the working protocol

(None recorded as of 2026-05-08. Any subsequent changes should be appended below with date, scope, and justification.)

Reformatted from `simulation-grid-plan.md` *on 2026-05-08 07:35 PDT. Source: ~/Dropbox/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/carryover-sensitivity/00-ademp-protocol.md.*